

The Light Plane: An Exact Horizon Map

Prime colors, positive geometry, recoverability,
and the consequences of a governing growth law

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Mathematical derivations prepared in collaboration with ChatGPT

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Abstract

We construct a normalized source state whose prime-occupation amplitudes encode the complete Möbius signs, a self-adjoint source Hamiltonian, and an explicit horizon map into the arithmetic field of the light-plane framework. The map preserves the full signed shell energy, including the terminal contribution. A single prime color becomes a signed logarithmic translation of the field at its exact reachback cutoff. Its source-side energy matrix is a constant multiple of $\min(a, b)$, so its complete singular spectrum and its inverse are elementary and exact. Field samples recover the retained arithmetic coefficients through a local second-difference law. A uniform phase transport estimate compares every Hamiltonian phase with the unshifted field, independently of horizon size. The proposed subpower growth law is retained as an explicit framework premise. Its consequences are derived as exact restrictions on horizon-mode populations, field amplitudes, weighted integrability, and phase windows.

Working scope. This continuation realizes the prime-color-to-field correspondence within the proposed light-plane programme. The constructions and identities below are proved directly. The growth law is the governing premise specified in Section 10; statements using it cite that premise. The higher structure is studied through these defined objects and their arithmetic images.

Three main formulas. With $Z = 15/\pi^2$ and $\Psi_n = \mu(n)/(n\sqrt{Z})$, the horizon map satisfies

$$\mathcal{P}_N \Psi = \phi_N, \quad \mathcal{P}_N^* \mathcal{P}_N = Z[\min(a, b)]_{a, b \leq N}.$$

Its energy modes have exact gains

$$\lambda_{j, N} = \frac{1}{4 \sin^2 \left(\frac{(2j-1)\pi}{4N+2} \right)}.$$

For source phase τ and $q(\tau) = \sqrt{1 + \tau^2} + |\tau|$,

$$q(\tau)^{-2} R(N) \leq R_N(\tau) \leq q(\tau)^2 R(N).$$

These formulas retain the complete signed arithmetic source.

Contents

1	Arithmetic field and its exact metric	3
2	A normalized source built from prime colors	4
3	The horizon map and its adjoint	4
4	A single prime color and its complete reachback	5
5	Exact horizon spectrum	6
6	Information retained and exact local recovery	7
7	The two tails and the terminal boundary	8
8	From Möbius colors to prime-power spectral colors	9
9	Hamiltonian phase and a uniform transport bound	9
10	The governing growth law and what it implies	10
11	The successor interaction in field coordinates	12
12	Dictionary, invariants, and reproducibility	12

1 Arithmetic field and its exact metric

For $N \in \mathbb{N}$, define

$$M_N(x) = \sum_{\substack{n \leq N \\ n \leq x}} \mu(n), \quad h_N(x) = \sum_{\substack{n \leq N \\ n \leq x}} \frac{\mu(n)}{n}.$$

Both functions vanish for $x < 1$ and become constant for $x \geq N$. Write $M(N) = M_N(N)$ and $h(N) = h_N(N)$. The energy is

$$R(N) = \sum_{a, b \leq N} \frac{\mu(a)\mu(b)}{\max(a, b)} = \int_1^N \frac{M(x)^2}{x^2} dx + \frac{M(N)^2}{N}. \quad (1)$$

Here and below $M(x) = \sum_{n \leq x} \mu(n)$ when no truncation is needed.

The field space is $\mathcal{E} = H^1(\mathbb{R})$ with inner product, conjugate-linear in the first variable,

$$\langle f, g \rangle_{\mathcal{E}} = \int_{\mathbb{R}} \left(\overline{f'} g' + \frac{1}{4} \overline{f} g \right) du.$$

Let

$$H_0 = -\partial_u^2 + \frac{1}{4}, \quad G_v(u) = e^{-|u-v|/2}.$$

As a differential operator on $L^2(\mathbb{R})$, H_0 has domain $H^2(\mathbb{R})$. It is positive and self-adjoint. In distributions, $H_0 G_v = \delta_v$: the derivative has jump -1 at v , and the homogeneous equation holds on both sides.

Proposition 1.1 (Reproducing metric). *For every $v, w \in \mathbb{R}$,*

$$\langle G_v, f \rangle_{\mathcal{E}} = f(v), \quad \langle G_v, G_w \rangle_{\mathcal{E}} = e^{-|v-w|/2}, \quad \|G_v\|_{\mathcal{E}} = 1.$$

Consequently, for any finite real or complex coefficients z_n ,

$$\phi_z(u) = \sum_{n \leq N} \frac{z_n}{\sqrt{n}} G_{\log n}(u) \implies \|\phi_z\|_{\mathcal{E}}^2 = \sum_{a, b \leq N} \frac{\overline{z_a} z_b}{\max(a, b)}. \quad (2)$$

Proof. Integration by parts gives the reproducing identity first on smooth compactly supported functions, and continuity extends it to $H^1(\mathbb{R})$. The remaining identities follow by substitution and $e^{-|\log a - \log b|/2} / \sqrt{ab} = 1 / \max(a, b)$. $\square$

The actual arithmetic field is $\phi_N = \phi_{\mu}$, so

$$H_0 \phi_N = J_N, \quad J_N = \sum_{n \leq N} \frac{\mu(n)}{\sqrt{n}} \delta_{\log n}, \quad \|\phi_N\|_{\mathcal{E}}^2 = R(N).$$

Its full-line Fourier representation, with $\widehat{f}(k) = \int f(u) e^{-iku} du$, is

$$R(N) = \frac{1}{2\pi} \int_{\mathbb{R}} \frac{|D_N(\frac{1}{2} + ik)|^2}{k^2 + \frac{1}{4}} dk, \quad D_N(s) = \sum_{n \leq N} \mu(n) n^{-s}. \quad (3)$$

This follows from $\widehat{G}_0(k) = (k^2 + 1/4)^{-1}$ and Plancherel. The numerator is the complete finite signed sum.

2 A normalized source built from prime colors

Take the positive source Hilbert space $\mathcal{H} = \ell^2(\mathbb{N})$, with orthonormal basis $|n\rangle$. Unique prime factorization identifies $|n\rangle$ with the occupation list $(\nu_p(n))_p$. This is the integer prime-occupation construction used in the earlier Prime DNA work; all definitions needed here are explicit.

The normalization

$$Z = \sum_{n \geq 1} \frac{\mu(n)^2}{n^2} = \prod_p (1 + p^{-2}) = \frac{\zeta(2)}{\zeta(4)} = \frac{15}{\pi^2}$$

is finite. Define

$$\Psi = \frac{1}{\sqrt{Z}} \sum_{n \geq 1} \frac{\mu(n)}{n} |n\rangle, \quad \|\Psi\|_{\mathcal{H}} = 1. \quad (4)$$

In prime-occupation coordinates this is the convergent product state

$$\Psi = \bigotimes_p \frac{|0\rangle_p - p^{-1}|1\rangle_p}{\sqrt{1 + p^{-2}}}. \quad (5)$$

Expansion gives sign $(-1)^{|S|}$ and amplitude $\prod_{p \in S} p^{-1}$ for a finite prime set S . Repeated prime occupations have zero amplitude. Thus the product gives exactly $\mu(n)/n$, including the squarefree rule and every multiplicative sign. Convergence follows from $\sum_p p^{-2} < \infty$.

The source Hamiltonian is

$$\mathcal{H}_{\text{src}}|n\rangle = (\log n)|n\rangle, \quad \mathcal{D}(\mathcal{H}_{\text{src}}) = \left\{ c : \sum_n (\log n)^2 |c_n|^2 < \infty \right\}. \quad (6)$$

It is a self-adjoint diagonal operator. Its unitary evolution is

$$U(\tau)c = (n^{-i\tau} c_n)_{n \geq 1}.$$

The state Ψ belongs to the domain of every fixed power of $\mathcal{H}_{\text{src}}$. Its norm and its source Hamiltonian expectation are conserved. In particular,

$$\langle \Psi, \mathcal{H}_{\text{src}} \Psi \rangle = \sum_p \frac{\log p}{p^2 + 1} < \infty.$$

This source Hamiltonian generates phases $\log n$. The field operator H_0 defines spatial response and the field metric. Their respective roles are kept separate throughout.

3 The horizon map and its adjoint

Let Π_N retain the source coordinates $1, \dots, N$. Define the finite-rank map $\mathcal{P}_N : \mathcal{H} \rightarrow \mathcal{E}$ by

$$(\mathcal{P}_N c)(u) = \sqrt{Z} \sum_{n \leq N} \sqrt{n} c_n G_{\log n}(u). \quad (7)$$

Each color coordinate has a specified image: $\mathcal{P}_N |n\rangle = \sqrt{Zn} G_{\log n}$ for $n \leq N$, and zero for $n > N$. Its action on the actual source is exact:

$$\mathcal{P}_N \Psi = \phi_N, \quad \mathcal{P}_N U(\tau) \Psi = \sum_{n \leq N} \frac{\mu(n) n^{-i\tau}}{\sqrt{n}} G_{\log n}. \quad (8)$$

The source normalization is compensated explicitly by the map. No arithmetic coefficient is changed.

Theorem 3.1 (Exact positive pullback). *For $f \in \mathcal{E}$,*

$$(\mathcal{P}_N^* f)_n = \begin{cases} \sqrt{Zn} f(\log n), & n \leq N, \\ 0, & n > N. \end{cases}$$

On the retained coordinates,

$$\mathbf{A}_N := \mathcal{P}_N^* \mathcal{P}_N = Z \mathbf{B}_N, \quad (\mathbf{B}_N)_{ab} = \min(a, b). \quad (9)$$

In particular, $R(N) = \langle \Psi, \mathbf{A}_N \Psi \rangle_{\mathcal{H}}$, with $\mathbf{A}_N$ extended by zero outside the retained coordinates.

Proof. The adjoint formula is the reproducing identity. Its matrix entries are $Z\sqrt{ab}e^{-|\log a - \log b|/2} = Z \min(a, b)$. Evaluating on $\Psi_a = \mu(a)/(a\sqrt{Z})$ gives $\mu(a)\mu(b) \min(a, b)/(ab) = \mu(a)\mu(b)/\max(a, b)$. $\square$

The identity explains how a conserved source norm can have a changing arithmetic readout: the horizon uses the positive observable $\mathbf{A}_N$, which depends on N . At fixed N , the Hamiltonian phase also changes the source's alignment with that observable. The conservation statement concerns $\|\Psi\|_{\mathcal{H}}$; the growth law concerns $\langle \Psi, \mathbf{A}_N \Psi \rangle$.

4 A single prime color and its complete reachback

Let $S_p|n\rangle = |pn\rangle$ be prime multiplication in the source space and let $T_a f(u) = f(u - a)$ be translation in the field space. Translation is unitary for the field metric. Put $\mathcal{P}_0 = 0$. Directly on every basis vector,

$$\boxed{\mathcal{P}_N S_p = \sqrt{p} T_{\log p} \mathcal{P}_{\lfloor N/p \rfloor}.} \quad (10)$$

Thus multiplication of a source coordinate has an exact geometric image, with its numerical horizon retained.

For a finite prime set $\mathcal{Q}$, define $\mu_{\mathcal{Q}}(n) = \mu(n)$ when every prime factor of n lies in $\mathcal{Q}$, and zero otherwise. This includes $\mu_{\mathcal{Q}}(1) = 1$. Let $\phi_{\mathcal{Q}, N}$ be its field as in (2), with $\phi_{\mathcal{Q}, 0} = 0$.

Theorem 4.1 (Prime-color action). *If $p \notin \mathcal{Q}$ and $K = \lfloor N/p \rfloor$, then*

$$\boxed{\phi_{\mathcal{Q} \cup \{p\}, N} = \phi_{\mathcal{Q}, N} - p^{-1/2} T_{\log p} \phi_{\mathcal{Q}, K}.} \quad (11)$$

Writing $R_{\mathcal{Q}}(N) = \|\phi_{\mathcal{Q}, N}\|_{\mathcal{E}}^2$, its full energy update is

$$\begin{aligned} R_{\mathcal{Q} \cup \{p\}}(N) &= R_{\mathcal{Q}}(N) + p^{-1} R_{\mathcal{Q}}(K) \\ &\quad - 2p^{-1/2} \operatorname{Re} \langle \phi_{\mathcal{Q}, N}, T_{\log p} \phi_{\mathcal{Q}, K} \rangle_{\mathcal{E}}. \end{aligned} \quad (12)$$

Proof. Each squarefree product using p is uniquely pm , with m a squarefree product from $\mathcal{Q}$. Its sign is $-\mu_{\mathcal{Q}}(m)$, its weight is $(pm)^{-1/2}$, and its location is $\log p + \log m$. The restriction $pm \leq N$ is precisely $m \leq K$. Splitting the field sum proves (11). Squaring in the actual metric, using translation invariance and retaining the mixed inner product, proves (12). $\square$

This is a complete reachback operation: translation by $\log p$, signed coefficient $-p^{-1/2}$, and the inherited field through $\lfloor N/p \rfloor$. All pulses and both spatial tails of that inherited field are translated together.

On field families $\mathbf{f} = (f_N)_{N \geq 0}$ with $f_0 = 0$, define

$$(\mathcal{D}_p \mathbf{f})_N = p^{-1/2} T_{\log p} f_{\lfloor N/p \rfloor}.$$

These actions commute, since

$$(\mathcal{D}_p \mathcal{D}_q \mathbf{f})_N = (pq)^{-1/2} T_{\log(pq)} f_{\lfloor N/(pq) \rfloor}.$$

For the base family $g_N = G_0$ when $N \geq 1$ and $g_0 = 0$, this gives

$$\phi_N = \left[\prod_{p \leq N} (I - \mathcal{D}_p) \mathbf{g} \right]_N. \quad (13)$$

Expansion is exactly the full Möbius field. The cutoff removes only products exceeding N ; no estimate or independent-sign assumption is used. The source Hamiltonian gives each prime action its frequency: $[\mathcal{H}_{\text{src}}, S_p] = (\log p) S_p$ on finite source vectors.

5 Exact horizon spectrum

Let $\mathbf{C}_N$ be the lower triangular matrix with ones on and below the diagonal. Then $\mathbf{B}_N = \mathbf{C}_N \mathbf{C}_N^\top$. Its inverse is

$$\mathbf{L}_N = \mathbf{B}_N^{-1} = \begin{pmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 2 & -1 \\ & & & -1 & 1 \end{pmatrix}, \quad (14)$$

with $\mathbf{L}_1 = (1)$.

Theorem 5.1 (Complete singular decomposition). *For $j = 1, \dots, N$, set*

$$\theta_{j,N} = \frac{(2j-1)\pi}{2N+1}, \quad v_{j,N}(n) = \frac{2}{\sqrt{2N+1}} \sin(n\theta_{j,N}), \quad \lambda_{j,N} = \frac{1}{4 \sin^2(\theta_{j,N}/2)}.$$

The vectors $v_{j,N}$ form an orthonormal basis and $\mathbf{B}_N v_{j,N} = \lambda_{j,N} v_{j,N}$. The singular values of $\mathcal{P}_N$ are $\sqrt{Z \lambda_{j,N}}$.

Proof. The inverse of $\mathbf{C}_N$ is first difference, with diagonal 1 and subdiagonal -1 , giving (14). Its eigenproblem is $2v_n - v_{n-1} - v_{n+1} = \lambda^{-1} v_n$, with boundaries $v_0 = 0$ and $v_{N+1} = v_N$. Substitution of $v_n = \sin(n\theta)$ gives inverse eigenvalue $2 - 2 \cos \theta$ and boundary condition $\cos((2N+1)\theta/2) = 0$. These give the displayed N distinct angles. Their squared sine norms are $(2N+1)/4$; symmetry gives orthogonality. The singular values follow from (9). $\square$

Define the actual signed mode amplitudes

$$b_{j,N} = \sum_{n \leq N} \frac{\mu(n)}{n} v_{j,N}(n). \quad (15)$$

Then the entire energy has the positive resolution

$$R(N) = \sum_{j=1}^N \lambda_{j,N} |b_{j,N}|^2. \quad (16)$$

Every signed interaction is retained inside the amplitudes $b_{j,N}$. There is no assumed orthogonality of the original arithmetic pulses. The new modes are obtained by diagonalizing their actual metric.

The field modes $w_{j,N} = \mathcal{P}_N v_{j,N} / \sqrt{Z \lambda_{j,N}}$ are orthonormal in $\mathcal{E}$, and

$$\phi_N = \sum_j \sqrt{\lambda_{j,N}} b_{j,N} w_{j,N}.$$

In particular,

$$\|\mathcal{P}_N\|^2 = \frac{Z}{4 \sin^2(\pi/(4N+2))} \sim \frac{4Z}{\pi^2} N^2.$$

For fixed j , $\lambda_{j,N} \sim 4N^2/((2j-1)^2\pi^2)$. These are the gains of the horizon observable $\mathbf{B}_N$; their indices label its sine modes.

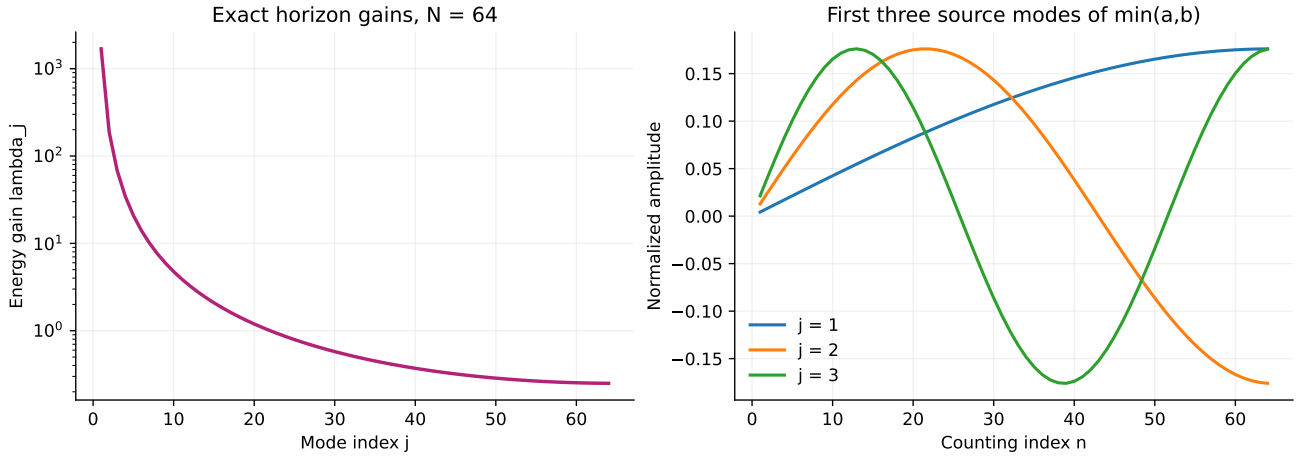


Figure 1: The exact gains and first three orthonormal modes of $\mathbf{B}_{64}$. The high gain of low-index modes makes the source's signed overlap with them quantitatively important.

6 Information retained and exact local recovery

The map has kernel $(I - \Pi_N)\mathcal{H}$ and is injective on $\Pi_N\mathcal{H}$. Indeed, applying H_0 to a vanishing finite field gives a sum of delta distributions at distinct points, forcing each retained coefficient to vanish.

There is also recovery from only N field samples. Set

$$w_n = \sqrt{n} \phi_N(\log n), \quad n = 1, \dots, N, \quad w_0 = 0.$$

Directly from the kernel,

$$w_n = \sum_{a \leq N} \frac{\mu(a)}{a} \min(n, a).$$

Thus $w = \mathbf{B}_N(\mu(a)/a)_{a \leq N}$. Applying $\mathbf{L}_N$ proves the following.

Theorem 6.1 (Local sample inversion). *For $1 \leq n < N$,*

$$\mu(n) = n(2w_n - w_{n-1} - w_{n+1}), \quad \mu(N) = N(w_N - w_{N-1}). \quad (17)$$

For $N = 1$, the terminal formula gives $\mu(1) = w_1$. The same identities hold for any finite source coefficients in place of $\mu(n)$.

The scalar energy alone compresses this information; the N samples retain the N coefficients exactly. Recovery is an algebraic statement about exact samples. Its numerical sensitivity depends on the matrix and the observation errors.

Two particularly simple observations are

$$\phi_N(0) = h(N), \quad \phi_N(\log N) = M(N)/\sqrt{N}.$$

Away from pulses, value and slope recover the two cumulative quantities:

$$M_N(e^u) = e^{u/2}(\phi_N(u)/2 - \phi'_N(u)), \quad (18)$$

$$h(N) - h_N(e^u) = e^{-u/2}(\phi_N(u)/2 + \phi'_N(u)). \quad (19)$$

At a source, the derivative jump is $[\phi'_N]_{\log n} = -\mu(n)/\sqrt{n}$.

7 The two tails and the terminal boundary

On the full line define

$$F_N(u) = e^{-u/2}M_N(e^u), \quad B_N(u) = e^{u/2}(h(N) - h_N(e^u)).$$

Then $\phi_N = F_N + B_N$ and, almost everywhere, $\phi'_N = (B_N - F_N)/2$. Hence

$$|\phi'_N|^2 + \frac{1}{4}|\phi_N|^2 = \frac{1}{2}|F_N|^2 + \frac{1}{2}|B_N|^2. \quad (20)$$

Both square integrals equal $R(N)$. The first gives (1); the second gives the exact harmonic-tail identity

$$\boxed{R(N) = h(N)^2 + \sum_{k=1}^{N-1} (h(N) - h(k))^2.} \quad (21)$$

This also follows by expanding the integral $\int_0^\infty (h(N) - h_N(x))^2 dx$ and using $\min(a, b)/(ab) = 1/\max(a, b)$. Equal shares refer to the two positive contributions in the completed field's density (20); the separate channels have jumps.

The exterior fields are explicit:

$$\phi_N(u) = \begin{cases} h(N)e^{u/2}, & u < 0, \\ M(N)e^{-u/2}, & u > \log N. \end{cases}$$

They carry energies $h(N)^2/2$ and $M(N)^2/(2N)$ respectively. Thus

$$R(N) = \int_0^{\log N} \left(|\phi'_N|^2 + \frac{1}{4}|\phi_N|^2 \right) du + \frac{1}{2}h(N)^2 + \frac{M(N)^2}{2N}.$$

The importance of the historical terminal is visible in the matrix. Removing $M(N)^2/N$ from (1) replaces the kernel by

$$K_{ab}^{\text{hist}} = \frac{1}{\max(a, b)} - \frac{1}{N},$$

and replaces its weighted source matrix by $Z(\min(a, b) - ab/N)$. The last row and column then vanish. The full terminal restores the positive rank-one matrix $Z(ab/N)$. This is exactly the term needed to retain the last source coefficient.

8 From Möbius colors to prime-power spectral colors

The source also determines the prime-power wave in the light-plane paper. The exact finite coefficient identity is

$$\Lambda(n) = \sum_{d|n} \mu(d) \log(n/d). \quad (22)$$

For $n = p^k$, the nonzero divisor contributions give $\log p$. If n has at least two distinct prime factors, expansion over their subsets cancels both the constant and the linear logarithmic terms. For $n = 1$, the sum is zero. This proves (22) directly.

Consequently the recovered source through N determines, for every specified real window $\omega(n)$, the finite spectral readout

$$\mathcal{W}_{N,\omega}(\tau) = -2 \sum_{n \leq N} \omega(n) \frac{\Lambda(n)}{\sqrt{n}} \cos(\tau \log n). \quad (23)$$

Taking $\omega(n) = 1 - \log n / \log N$ for $N > 1$ reproduces the finite wave used in Section 7 of *The Light Plane*. Thus the source, the Möbius field, and the prime-power wave have a fully specified finite correspondence.

In the absolute-convergence half-plane $\Re s > 1$, this relationship is also expressed by

$$D(s) = \prod_p (1 - p^{-s}) = \frac{1}{\zeta(s)}, \quad \frac{D'(s)}{D(s)} = \sum_{p,k \geq 1} (\log p) p^{-ks}.$$

Finite product and numerical cutoffs are distinct. For the prime set $\mathcal{Q}_N = \{p : p \leq N\}$,

$$\prod_{p \in \mathcal{Q}_N} (1 - p^{-s}) = D_N(s) + \sum_{\substack{n > N \\ p|n \Rightarrow p \in \mathcal{Q}_N}} \mu(n) n^{-s}.$$

The second sum is finite because only squarefree products contribute. This displayed remainder keeps the products beyond the numerical horizon explicit. Identity (22) requires no replacement of one cutoff by the other.

9 Hamiltonian phase and a uniform transport bound

Define

$$R_N(\tau) = \|\mathcal{P}_N U(\tau) \Psi\|_{\mathcal{E}}^2, \quad R_N(0) = R(N).$$

The full interaction is

$$R_N(\tau) = \sum_{n \leq N} \frac{\mu(n)^2}{n} + 2 \sum_{a < b \leq N} \frac{\mu(a)\mu(b)}{b} \cos\left(\tau \log \frac{b}{a}\right). \quad (24)$$

It follows that the long phase average, at each fixed N , is

$$\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T R_N(\tau) d\tau = \sum_{n \leq N} \frac{\mu(n)^2}{n}.$$

All unequal-index frequencies have nonzero frequency, so their finite averages tend to zero.

Theorem 9.1 (Uniform phase transport). *Let $q(\tau) = \sqrt{1 + \tau^2} + |\tau|$. For every N and real τ ,*

$$\boxed{q(\tau)^{-2} R(N) \leq R_N(\tau) \leq q(\tau)^2 R(N)}. \quad (25)$$

The same estimate holds for any finite complex arithmetic source.

Proof. Replacing the source coefficient $\mu(n)$ by $\mu(n)n^{-i\tau}$ in (3), then changing variables, gives

$$R_N(\tau) = \frac{1}{2\pi} \int_{\mathbb{R}} \frac{|D_N(\frac{1}{2} + it)|^2}{(t - \tau)^2 + \frac{1}{4}} dt.$$

For every real t ,

$$q(\tau)^{-2} \leq \frac{t^2 + \frac{1}{4}}{(t - \tau)^2 + \frac{1}{4}} \leq q(\tau)^2.$$

To verify the constants, put $y = t - \tau$ and express the ratio as the squared-norm ratio of the shear $(y, 1/2) \mapsto (y + \tau, 1/2)$. The squared singular values of the matrix $\begin{pmatrix} 1 & 2\tau \\ 0 & 1 \end{pmatrix}$ are $q(\tau)^2$ and $q(\tau)^{-2}$. Integrating against the nonnegative density $|D_N(1/2 + it)|^2/(t^2 + 1/4)$ proves the claim. $\square$

This calculation transports the full signed numerator. The bound's constant is independent of N and of the coefficient pattern. The source's unitary phase motion therefore has a controlled, generally nonconstant, energy readout at the horizon.

For example, at $N = 2$,

$$R_2(\tau) = \frac{3}{2} - \cos(\tau \log 2).$$

The source norm remains one, while its projected energy ranges from $1/2$ to $5/2$.

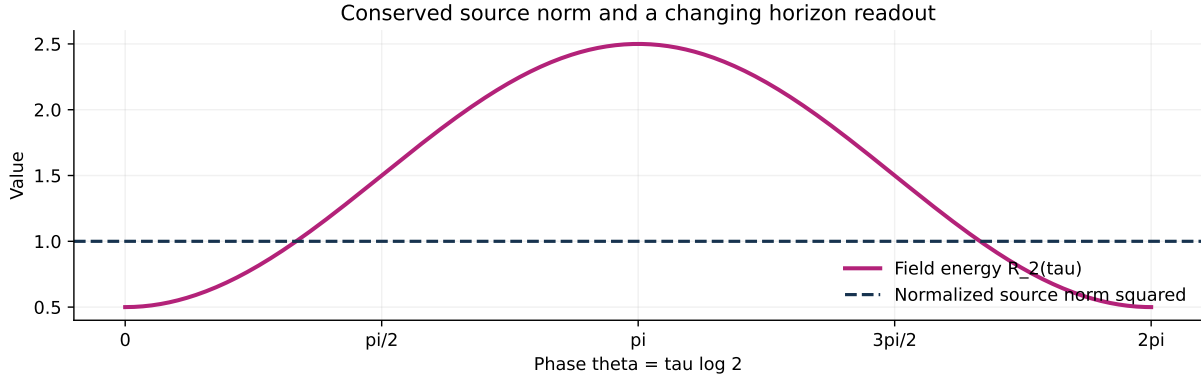


Figure 2: An exact two-source example. The phase parameter changes the complete interference term while the normalized source norm is conserved.

10 The governing growth law and what it implies

Framework premise 10.1 (G: growth in the chosen arithmetic image). *For every $\varepsilon > 0$ there is a constant C_ε independent of N such that*

$$R(N) = \|\mathcal{P}_N \Psi\|_{\mathcal{E}}^2 \leq C_\varepsilon N^\varepsilon. \quad (26)$$

This is the working growth law supplied by the light-plane framework. The following deductions use it as stated.

Corollary 10.2 (Exact mode-population allowance). *For $1 \leq J \leq N$,*

$$\sum_{j=1}^N \lambda_{j,N} |b_{j,N}|^2 \leq C_\varepsilon N^\varepsilon, \quad (27)$$

$$\begin{aligned} \sum_{j=1}^J |b_{j,N}|^2 &\leq 4C_\varepsilon N^\varepsilon \sin^2\left(\frac{(2J-1)\pi}{4N+2}\right) \\ &\leq \frac{C_\varepsilon \pi^2 (2J-1)^2 N^\varepsilon}{(2N+1)^2}. \end{aligned} \quad (28)$$

The allowance is $O_\varepsilon(J^2 N^{-2+\varepsilon})$. Dividing $\sum_{j \leq J} |b_{j,N}|^2$ by Z gives the source probability in these modes.

Proof. Equation (27) is (16) and Premise G. The gains decrease with j , so the first J terms of the energy dominate $\lambda_{j,N} \sum_{j \leq J} |b_{j,N}|^2$. Use the exact inverse gain and $\sin x \leq x$. $\square$

Thus the premise has a concrete geometric meaning: as the horizon enlarges, the actual signed source has increasingly small total overlap with its high-gain, low-index modes. The statement concerns the combined color amplitudes in (15), retaining every interaction.

Corollary 10.3 (Field, terminal, and damping consequences). *Under Premise G,*

$$\sup_{u \in \mathbb{R}} |\phi_N(u)|^2 \leq C_\varepsilon N^\varepsilon, \quad (29)$$

$$|M(N)|^2 \leq C_\varepsilon N^{1+\varepsilon}, \quad (30)$$

$$\int_{\mathbb{R}} |\phi'_N|^2 du \leq C_\varepsilon N^\varepsilon, \quad \int_{\mathbb{R}} |\phi_N|^2 du \leq 4C_\varepsilon N^\varepsilon. \quad (31)$$

For every $\delta > 0$ the infinite cumulative state $F(u) = e^{-u/2} M(e^u)$ satisfies

$$\int_0^\infty e^{-2\delta u} |F(u)|^2 du < \infty. \quad (32)$$

Proof. Evaluation has norm one by Proposition 1.1. The terminal identity or $\phi_N(\log N) = M(N)/\sqrt{N}$ gives the second estimate, and the norm definition gives the next two. The terminal estimate at every integer implies $|F(u)|^2 \leq C_\eta e^{\eta u}$ for every $\eta > 0$. Choose $0 < \eta < 2\delta$ and integrate. In particular the integral is at most $C_\eta/(2\delta - \eta)$ after extending the integer bound to e^u by its floor. $\square$

Corollary 10.4 (Phase windows inherit the growth law). *For every fixed T ,*

$$\sup_{|\tau| \leq T} R_N(\tau) \leq q(T)^2 C_\varepsilon N^\varepsilon.$$

More generally, if $T(N) \geq 0$ and $1 + T(N) = N^{o(1)}$, then

$$\sup_{|\tau| \leq T(N)} R_N(\tau) = N^{o(1)}.$$

If $|\tau| \leq N^\kappa$ for a fixed $\kappa \geq 0$, the corresponding bound is $O_\varepsilon(N^{2\kappa+\varepsilon})$.

Proof. Apply Theorem 9.1 and $q(\tau) \leq 1 + 2|\tau|$. The subpower-window assertion follows by allocating any requested exponent between the energy premise and the phase factor. $\square$

All these conclusions follow from the same premise and the constructed map. In particular, the phase analysis introduces no additional arithmetic growth assumption.

11 The successor interaction in field coordinates

Adding the next source gives

$$\phi_n - \phi_{n-1} = \frac{\mu(n)}{\sqrt{n}} G_{\log n}.$$

The reproducing identity and $\phi_{n-1}(\log n) = M(n-1)/\sqrt{n}$ yield

$$\boxed{R(n) - R(n-1) = \frac{\mu(n)^2 + 2\mu(n)M(n-1)}{n}.} \quad (33)$$

This is the existing signed successor law in the exact field metric. The entire inherited source enters through the value of the field at the new source location. Its self-energy and its inherited interaction are both present.

For the phase-shifted source $M_\tau(n-1) = \sum_{a < n} \mu(a)a^{-i\tau}$, the same computation gives

$$R_n(\tau) - R_{n-1}(\tau) = \frac{\mu(n)^2 + 2 \operatorname{Re}\{\mu(n)n^{-i\tau}\overline{M_\tau(n-1)}\}}{n}.$$

The horizon's source support increases, while this signed energy increment may have either sign. That behavior is compatible with its positive total field energy.

12 Dictionary, invariants, and reproducibility

Object	Definition and role
Prime colors	Prime occupations of $ n\rangle$; the product (5) gives every actual Möbius amplitude.
Prime reachback	The exact action (11): the inherited field at $\lfloor N/p \rfloor$ translated by $\log p$ with coefficient $-1/\sqrt{p}$. Its complete interaction is (12).
Positive source geometry	$\ell^2(\mathbb{N})$ with its usual positive inner product and the normalized state (4).
Source Hamiltonian	$\mathcal{H}_{\text{src}} n\rangle = (\log n) n\rangle$; it generates the source phases and conserves its norm and energy expectation.
Horizon map	$\mathcal{P}_N$ in (7), including both the retained coordinate cutoff and the explicit weight converting the source to the arithmetic field.
Field geometry	$H^1(\mathbb{R})$ with norm $\int(\phi' ^2 + \phi ^2/4)$; its reproducing kernel is G_v .
Spatial response	$H_0\phi_N = J_N$; H_0 is the positive static field operator in logarithmic coordinates.
Horizon energy modes	The sine eigenvectors of $\min(a, b)$ with gains $\lambda_{j,N}$ from Theorem 5.1.
Spectral prime-power colors	The finite wave (23), recovered from the same source through (22).
Compression	Coordinates above N are outside the selected readout; the retained N amplitudes are exactly recoverable from N samples. The energy alone is a scalar summary.
Growth premise	Premise G, transported into mode allowances and phase windows by Corollaries 10.2–10.4.

The exact invariants under the change from source coefficients to the field description are the full signed quadratic energy, the retained coefficient data, the two cumulative boundary readings, and the

successor interaction. Source phase evolution separately conserves the source norm and its Hamiltonian expectation. The horizon readout transforms under the phase bound in Theorem 9.1.

The particular square-summable source normalization is optional. For any $\alpha > 1/2$, set

$$Z_\alpha = \sum_n \mu(n)^2 n^{-2\alpha}, \quad (\Psi_\alpha)_n = \mu(n) n^{-\alpha} / \sqrt{Z_\alpha}, \quad \mathcal{P}_{N,\alpha}|n\rangle = \sqrt{Z_\alpha} n^{\alpha-1/2} G_{\log n} \quad (n \leq N).$$

Then $\mathcal{P}_{N,\alpha}\Psi_\alpha = \phi_N$ and $\mathcal{P}_{N,\alpha}U(\tau)\Psi_\alpha = \phi_{N,\tau}$ exactly. These specified arithmetic readouts are independent of the compensating source-normalization choice.

In light units one may set $\tau = ct/\ell_0$ for a chosen length ℓ_0 . The phase formulas then use the dimensionless quantity ct/ℓ_0 . The map, the finite horizon gains, and the unshifted energy remain the same mathematical objects.

Reproduction. The accompanying script `figures_and_checks.py` generates both figures and checks the finite matrix, historical-energy, harmonic-tail, mode-energy, sample-inversion, and phase-bound formulas at $N = 1, 2, 3, 8, 32, 64$. The proofs in this manuscript hold for every finite N ; the finite computations check their implementation. Run the script and compile this source twice with `pdflatex`.

Source relationship. The field and shell definitions continue *The Light Plane*, Sections 5–7, and the field continuation supplied on October 5, 2026. The normalized prime-occupation state and diagonal source Hamiltonian use the architecture recorded in *Prime-State All-Pass Geometry and the Observability Boundary*, Prime DNA Build v32 (August 25, 2026). The horizon pullback, the complete prime-color reachback, the explicit mode resolution, the local sample inverse, and the phase-transport deductions are presented here with self-contained proofs.